



Yuweien / SiWeaver

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SiWeaver / README.md



Yuweien Update README.md

e7fdefc · 4 days ago



284 lines (170 loc) · 4.97 KB

Preview

Code

Blame

Raw



MindWeaver

A dialogue-driven interface for real-time structure visualization.

MindWeaver is a concept for an AI-native cognitive tool that externalizes thinking structures as they emerge.

This project explores a new interface paradigm where **thinking and structure generation happen simultaneously**, enabling individuals and teams to visualize and navigate complex ideas as they emerge.

Vision

Human thinking is not linear.

Ideas emerge through:

- dialogue
- writing
- discussion
- exploration

However, most tools capture thinking only as **linear text**, forcing structure to be added later.

This project proposes a new paradigm:

Thinking → Structure emerges in real time

Instead of writing first and structuring later, the system continuously externalizes the structure of thought.

Core Interface Concept

The interface is divided into two synchronized spaces.

Structure Space (Left Panel)	Thinking Space (Right Panel)
visual structure	text / dialogue / voice
concept map	writing
argument map	AI conversation
network graph	speech transcription

Thinking Space (Right Panel)

Where ideas are generated.

Possible modes:

- AI dialogue
- free writing
- speech-to-text discussion
- collaborative conversation

Structure Space (Left Panel)

Where the system externalizes thinking structures in real time.

Possible visualizations:

- outline tree
- mind map
- concept network
- argument map

Core Principle

Thinking produces structure.

input → semantic parsing → structural mapping → visualization



Three Core Use Cases

###1. Individual Thinking

Dialogue-driven thinking.

Users explore ideas through conversation with an AI or a simulated thinking partner.

dialogue
↓
concept extraction
↓
structure visualization



This allows users to see how their ideas evolve over time.

###2. Writing & Knowledge Construction

Users write normally, while the system generates structural representations of the text.

Examples:

- article writing
- academic research
- product design documents

text
↓
topic detection
↓
outline generation



↓

concept mapping

The system reveals hidden structure inside long-form writing

###3. Collaborative Discussion

For meetings and group discussions.

Before the meeting, an **agenda** is uploaded, which forms the structural backbone of the discussion.

agenda → core network



discussion → semantic mapping

visualization → discussion navigation

This enables participants to see:

- how ideas relate to the agenda
- when discussion drifts off-topic
- emerging new topics

##Structural Anchoring via Agenda

In collaborative scenarios, the agenda acts as a high-weight structural anchor.

Example:

```
Meeting Topic
├─ Agenda Item 1
├─ Agenda Item 2
└─ Agenda Item 3
```



Incoming discussion is mapped onto this structure.

Possible outcomes:

- Related to an agenda node
- Peripheral tangent

- New emerging topic

This allows real-time visualization of discussion dynamics.

##Visualization Modes

The system supports multiple structural views.

###Outline View

Hierarchical structure.

```
Topic
├─ Subtopic
├─ Argument
└─ Evidence
```



Best for writing and document design.

###Mind Map

Radial idea exploration.

Best for brainstorming and early-stage thinking.

####Network Graph

Relationship mapping between concepts.

Best for complex discussions and knowledge exploration.

##Conceptual Architecture

```
Input Layer
├─ dialogue
├─ writing
└─ speech

Processing Layer
├─ semantic parsing
├─ concept extraction
└─ relationship detection

Structure Layer
├─ topic hierarchy
├─ argument mapping
└─ concept graph
```



Visualization Layer

- |— outline
- |— mind map
- └— network graph

##Design Philosophy

This project is based on several principles.

1. Thinking is non-linear

Human cognition operates through networks of ideas.

2. Structure should emerge naturally

Users should not need to manually organize ideas during thinking.

3. Visualization enhances cognition

- Seeing thinking structures helps users:
- maintain logical coherence
- discover connections
- navigate complexity

4. Collective thinking benefits from shared structure

In group discussions, visualizing idea networks can prevent:

- topic drift
- repeated arguments
- unproductive tangents

Potential Applications

- research and academic thinking
- product design
- strategy meetings
- policy discussions

- knowledge exploration

Current Status

Concept exploration.

This repository documents the conceptual design of the system.

Future iterations may explore:

- prototype interfaces
- interaction models
- AI-assisted structure generation
- collaborative cognition tools

Author

Concept design by Yuwei Wang.

This project emerged from the need for tools that support dialogue-driven thinking and real-time cognitive structure visualization.